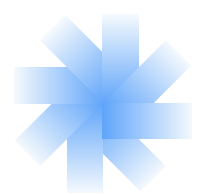


IAE MTP

Projet Crédit de risque

Presented by:

TEAM 2 : Ayoub, Ines,
Mohammed



Data Overview



Total

116 var

Après cleaning

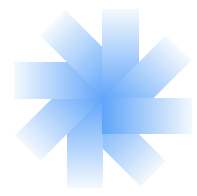
45 var

Total

31 286
ind

Après cleaning

9 710
ind



Cleaning

01

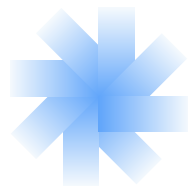
Suppression des
variables douteuses

02

imputation des NA

03

création de nouvelles
variables



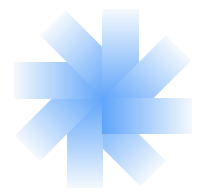
Nouvelles variables créées

$$\text{charges} = \text{pension alimentaire alimenataire verse} + \text{loyer} +$$
$$\text{charges loyer} + \text{mensualité credit} +$$
$$\text{charge_recurrentes_emp} + \text{charge courante_emp}$$

$$\text{horizon_retraite} = (\text{date_retraite_emp} -$$
$$\text{date_aboutisant_azur}) / 365,25$$

$$\text{recette} = \text{salaire emp} + \text{salaire_conjoint} + \text{apl} + \text{pension}$$
$$\text{alimentaire} + \text{allocation famille} + \text{pension invalidité} +$$
$$\text{rev_foncier_emp}$$

$$\text{horizon_rdv} = \text{date_rdv} - \text{date_aboutisant_azur}$$



XGboost

initial-70% train 30% test

```
Accuracy : 0.7881908685204256
Precision : 0.5333333333333333
Recall : 0.012944983818770227
F1-score : 0.02527646129541864
ROC-AUC : 0.6741343570869557
```

Matrice de confusion :

```
[[2288   7]
 [ 610   8]]
```

Classification report :

	precision	recall	f1-score	support
fait	0.79	1.00	0.88	2295
annuler client	0.53	0.01	0.03	618
accuracy			0.79	2913
macro avg	0.66	0.50	0.45	2913
weighted avg	0.74	0.79	0.70	2913

XGboost

equilibrer-70% train 30% test

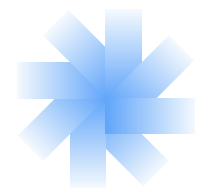
```
Matrice de confusion :
```

```
[[1212 1083]
 [ 150  468]]
```

```
Classification report :
```

	precision	recall	f1-score	support
fait	0.89	0.53	0.66	2295
annuler client	0.30	0.76	0.43	618
accuracy			0.58	2913
macro avg	0.60	0.64	0.55	2913
weighted avg	0.77	0.58	0.61	2913

```
ROC-AUC : 0.6782833090086089
```



XGboost

80% train 20% test

Matrice de confusion :

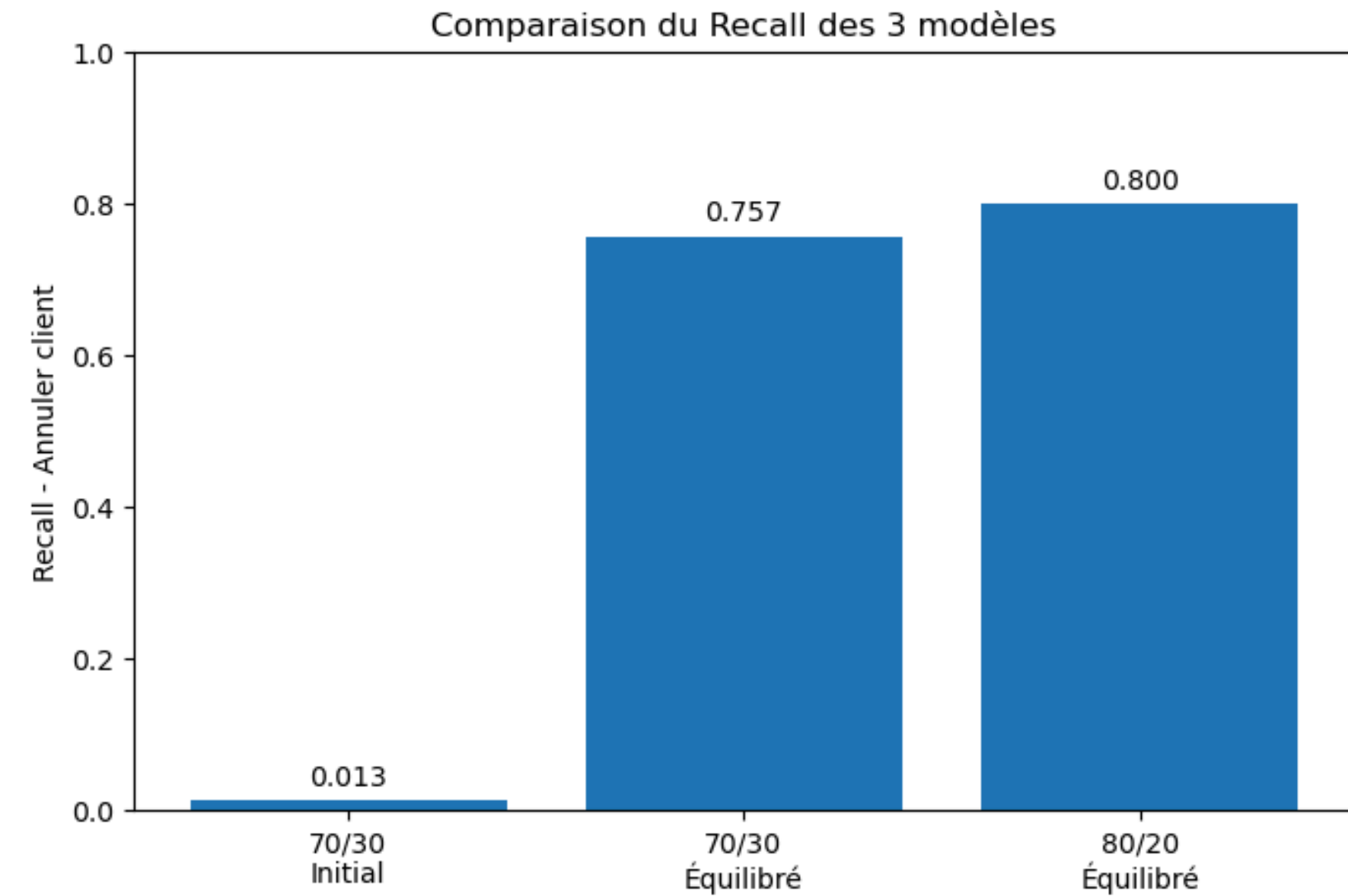
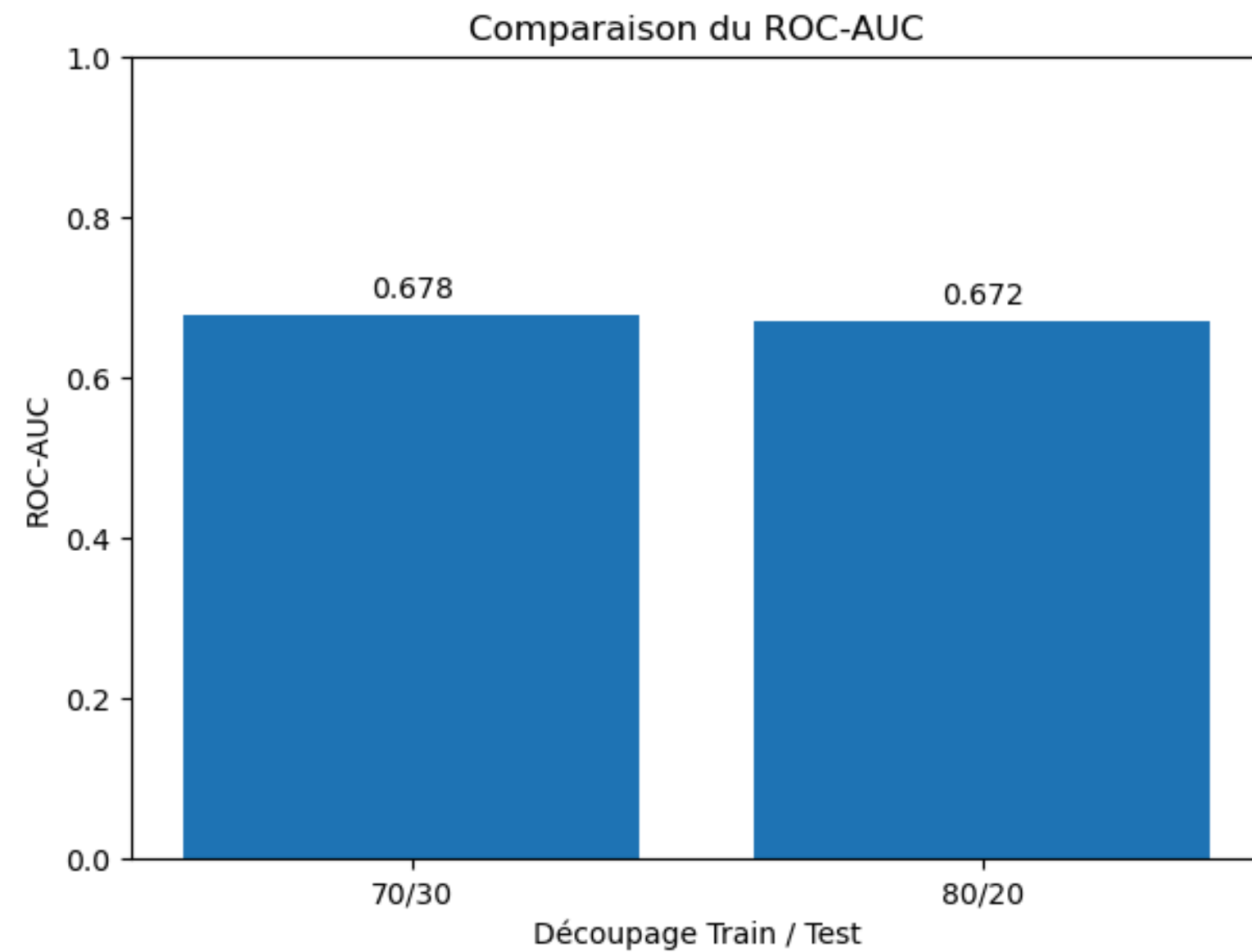
```
[[735 807]
 [ 80 320]]
```

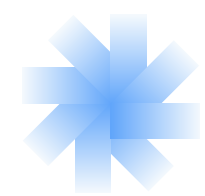
Classification report :

	precision	recall	f1-score	support
fait	0.90	0.48	0.62	1542
annuler client	0.28	0.80	0.42	400
accuracy			0.54	1942
macro avg	0.59	0.64	0.52	1942
weighted avg	0.77	0.54	0.58	1942

ROC-AUC : 0.6716107328145264

XGboost





Random forest

70% train et 30% test

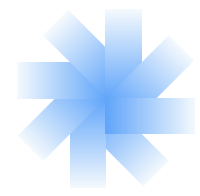
--- PERFORMANCES : RANDOM FOREST ---

	Prédit Fait (0)	Prédit Annulé (1)
Vrai Fait (0)	2501	0
Vrai Annulé (1)	285	127

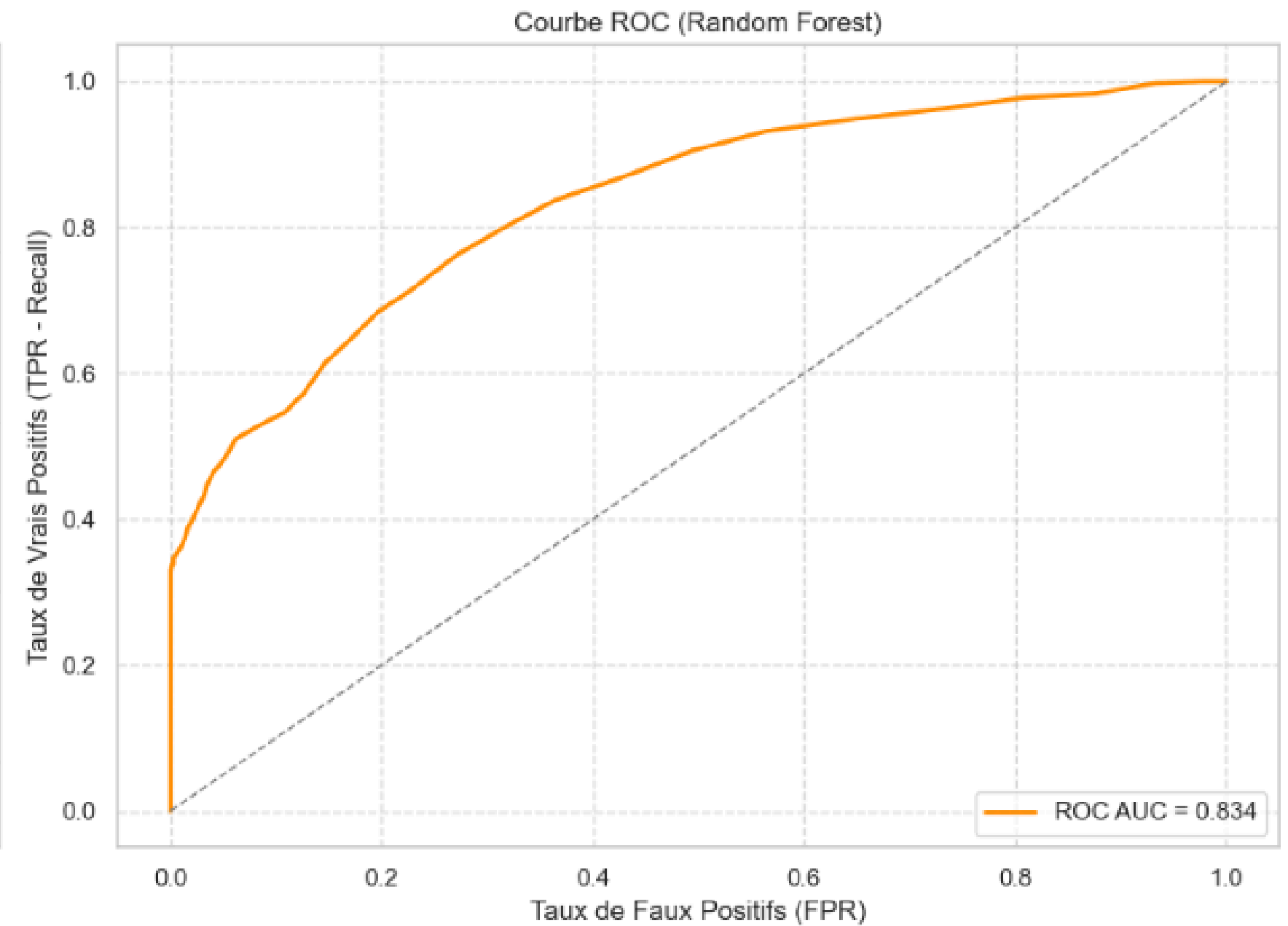
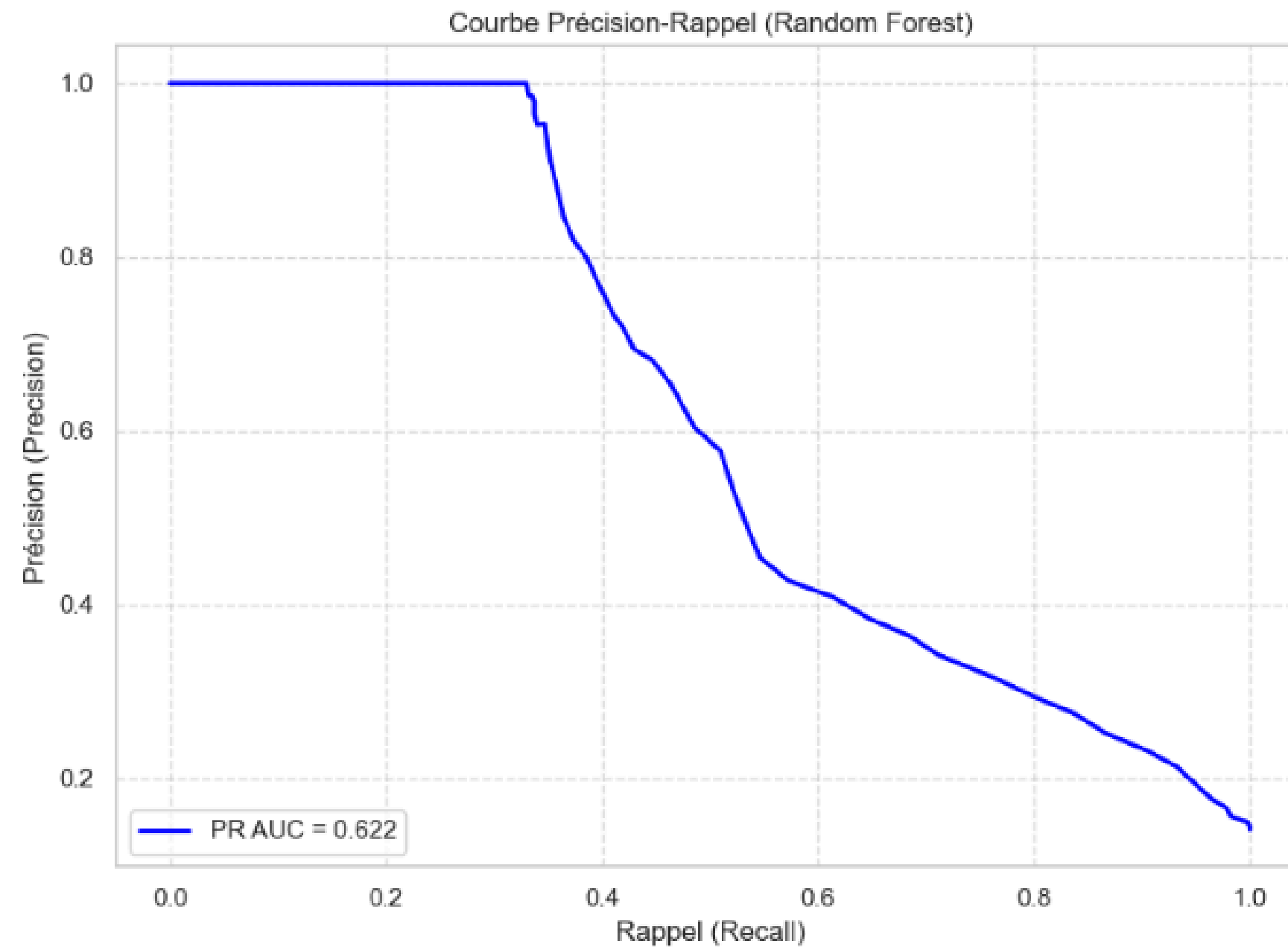
??

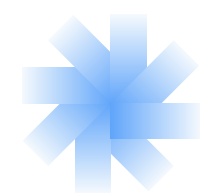
	precision	recall	f1-score	support
0	0.90	1.00	0.95	2501
1	1.00	0.31	0.47	412
accuracy			0.90	2913
macro avg	0.95	0.65	0.71	2913
weighted avg	0.91	0.90	0.88	2913

Score ROC-AUC : 0.834



Random forest

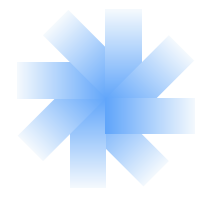




Regression Logistique

	Prédit Fait (0)	Prédit Annulé (1)		
Vrai Fait (0)	2355	146		
Vrai Annulé (1)	274	138		
	precision	recall	f1-score	support
0	0.90	0.94	0.92	2501
1	0.49	0.33	0.40	412
accuracy			0.86	2913
macro avg	0.69	0.64	0.66	2913
weighted avg	0.84	0.86	0.84	2913

Score ROC-AUC : 0.761

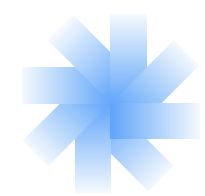


Arbre de décision

--- PERFORMANCES : ARBRE DE DÉCISION ---

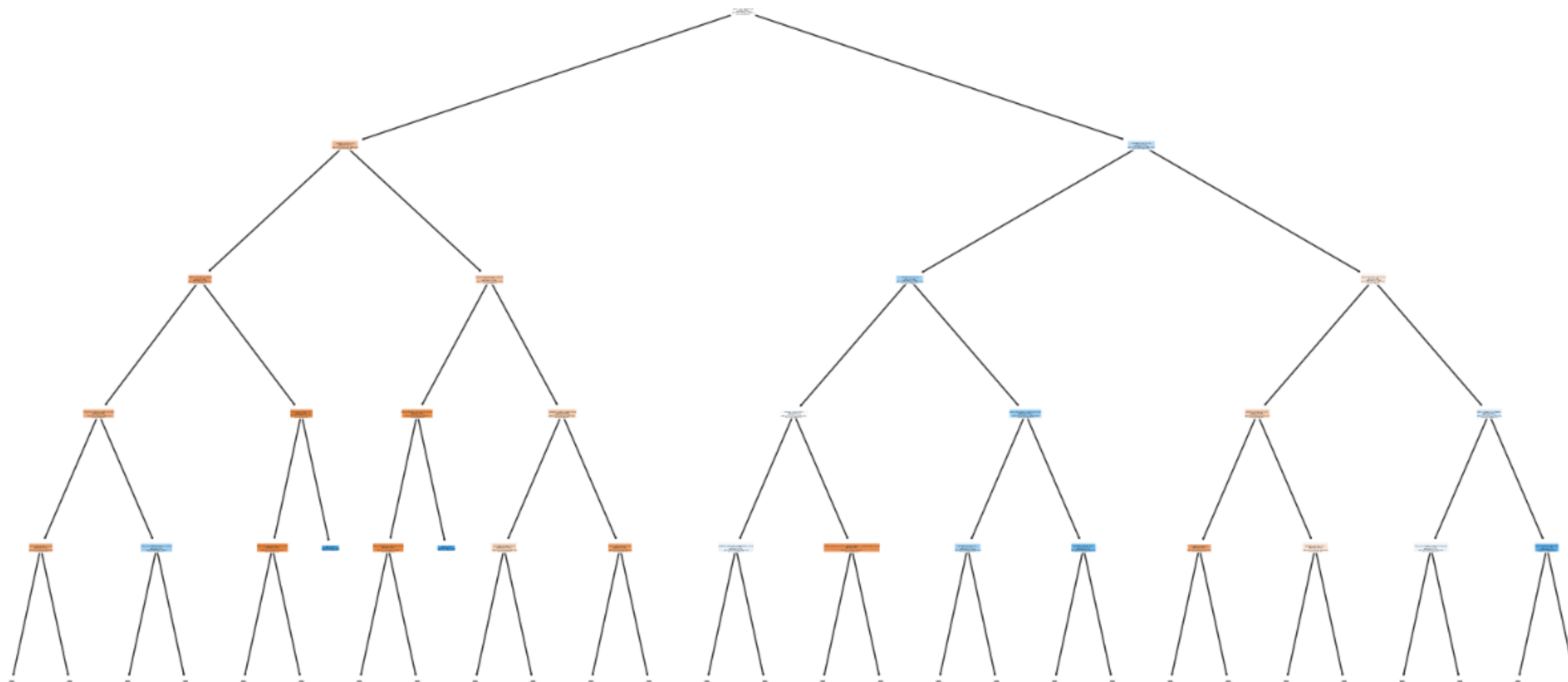
	Prédit Fait (0)	Prédit Annulé (1)			
Vrai Fait (0)	2251	250			
Vrai Annulé (1)	199	213			
	precision	recall	f1-score	support	
0	0.92	0.90	0.91	2501	
1	0.46	0.52	0.49	412	
accuracy			0.85	2913	
macro avg	0.69	0.71	0.70	2913	
weighted avg	0.85	0.85	0.85	2913	

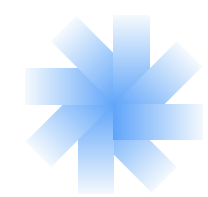
Score ROC-AUC : 0.709



Arbre de décision

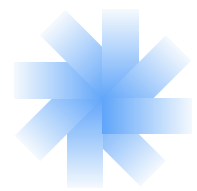
Visualisation de l'Arbre de Décision





Comparaison des modèles

Modèles	Precision/Rappel	F1-score	ROC-AUC
XGBOOST	0: 0.90 0.48 1: 0.28 0.80	1 : 0.42 0 : 0.62	0.671
Regression logistique	0: 0.90 0.95 1: 0.53 0.33	1 : 0.92 0 : 0.41	0.760
Forêt Aleatoire	0: 0.90 1.00 1: 1.00 0.29	1 : 0.95 0 : 0.45	0.834
Arbre de décision	0: 0.92 0.90 1: 0.46 0.52	1 : 0.91 0 : 0.49	0.709



**Thank
You**

